Predicting Vehicle Insurance Premiums with Machine Learning

Winter Term 2025
DATA 611 – Predictive Analytics
Group Project 2

Presented by Group 6:
Alvin Hui Tsz Chuen
Binh Nguyen
Judy Kurupakorn

Monday, February 24, 2025

UNIVERSITY OF CALGARY



Source: Our top five neighbourhoods in Calgary! (2024). discover.ucalgary.ca
DeepAI, Cyberpunk Generator

Introduction



Introduction

- Why the topic is important
  - **Understanding premium determinants** allows consumers to make informed decisions and insurers to price policies more accurately.
  - **Fair pricing benefits both individuals and businesses**, leading to improved financial planning and risk management.

UNIVERSITY OF CALGARY

Introduction

- Research Questions
  - How can we predict car insurance premiums using machine learning models?
  - What are the key factors influencing car insurance premiums?

- What algorithm will be used
  - **Random Forest (RF)**: Capturing complex, nonlinear relationships and handling interactions between variables effectively.
  - **Multiple Linear Regression (MLR)**: Interpretability and ability to quantify relationships between predictors and premiums.

UNIVERSITY OF CALGARY

Datasets



Datasets

- Vehicle Insurance Data
  - Mendeley Data
  - Obtained from the Ethiopian Insurance Corporation
  - 2014–2018 data
  - 16 columns and 508,499 rows
- Data cleaning
  - Missing values
  - Zero values
  - Feature engineering i.e. extract duration in days, flag claim status

sex	insurance_begin	insurance_end	effective_year	insurance_type	insured_value	premium	production_year
legal entity	2014-08-08	2015-08-07	08	commercial	3400000.00	26804.72	2007
legal entity	2017-08-08	2018-08-07	08	commercial	920000.0	12130.970	2010
male	2018-02-02	2019-02-01	12	private	164516.00	294.51	2018

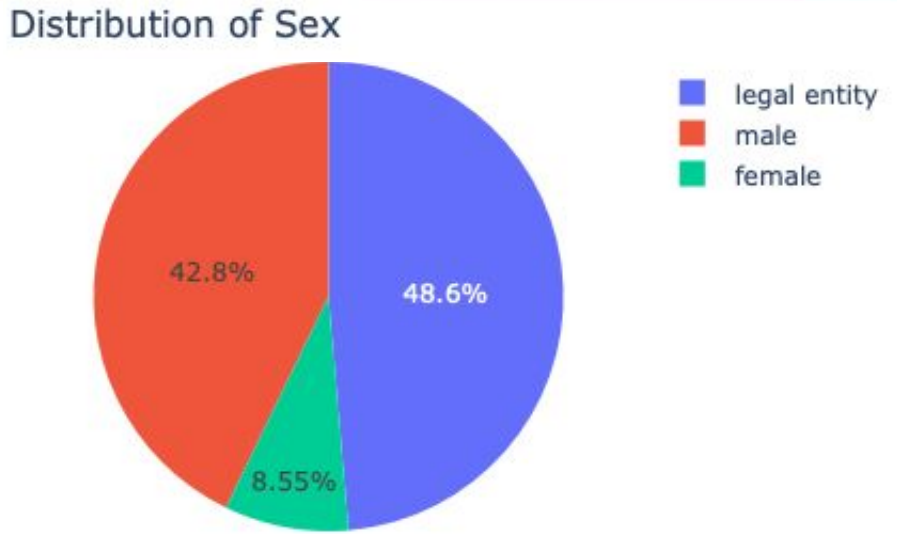
seats_number	carrying_capacity	vehicle_type	ccm_ton	manufacturer	usage	claim_paid
4.0	220.0	Truck	3153	NISSAN	General Cartage	NaN
7.0	0.0	Special construction	4164	TOYOTA	Ambulance	NaN
4.0	7.0	Pick-up	2400	TOYOTA	Private	NaN

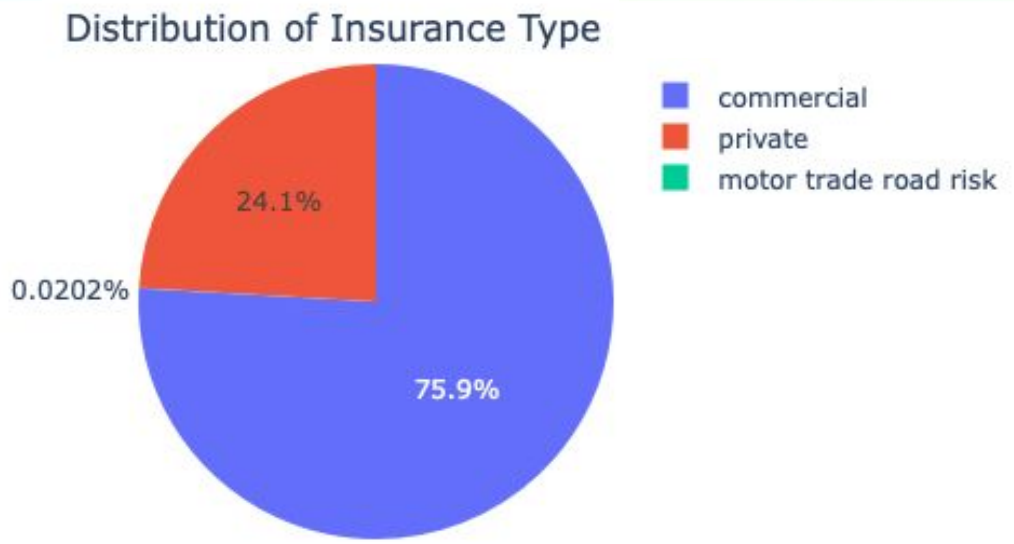
UNIVERSITY OF CALGARY

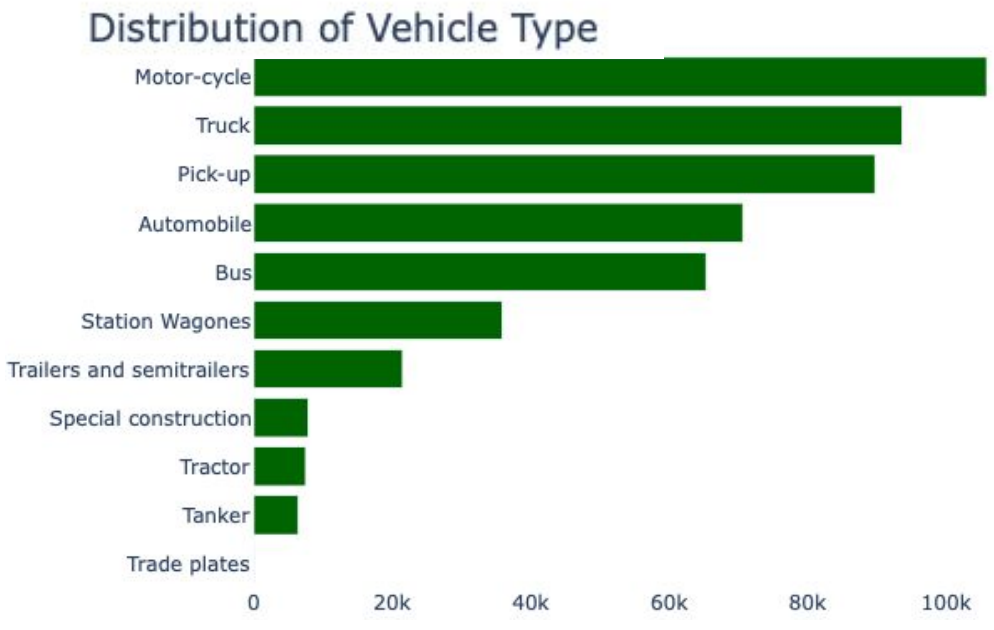
Data Visualization

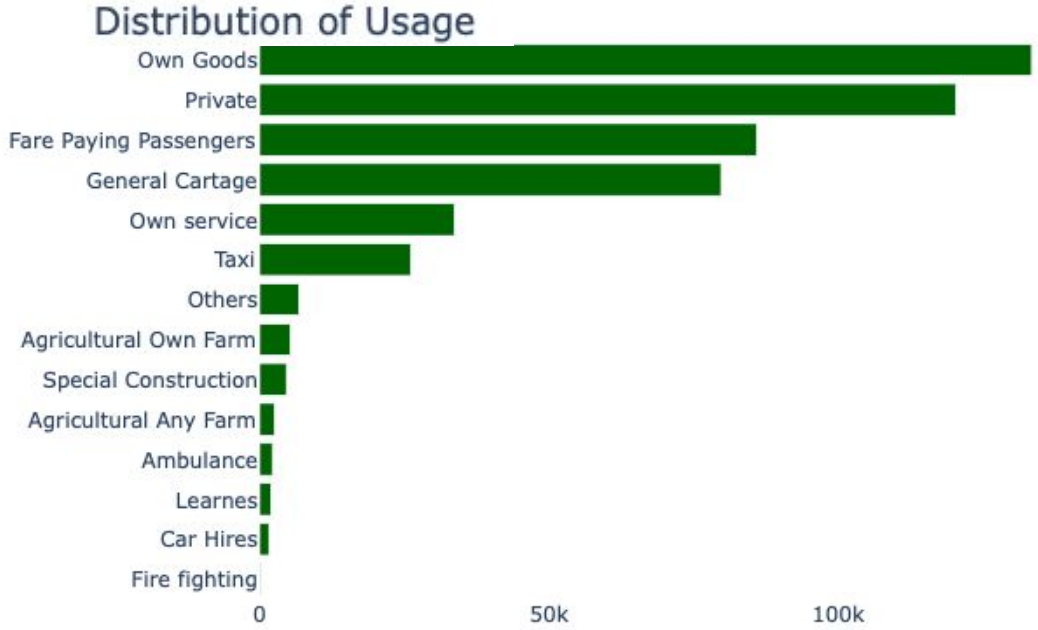


Data Visualization



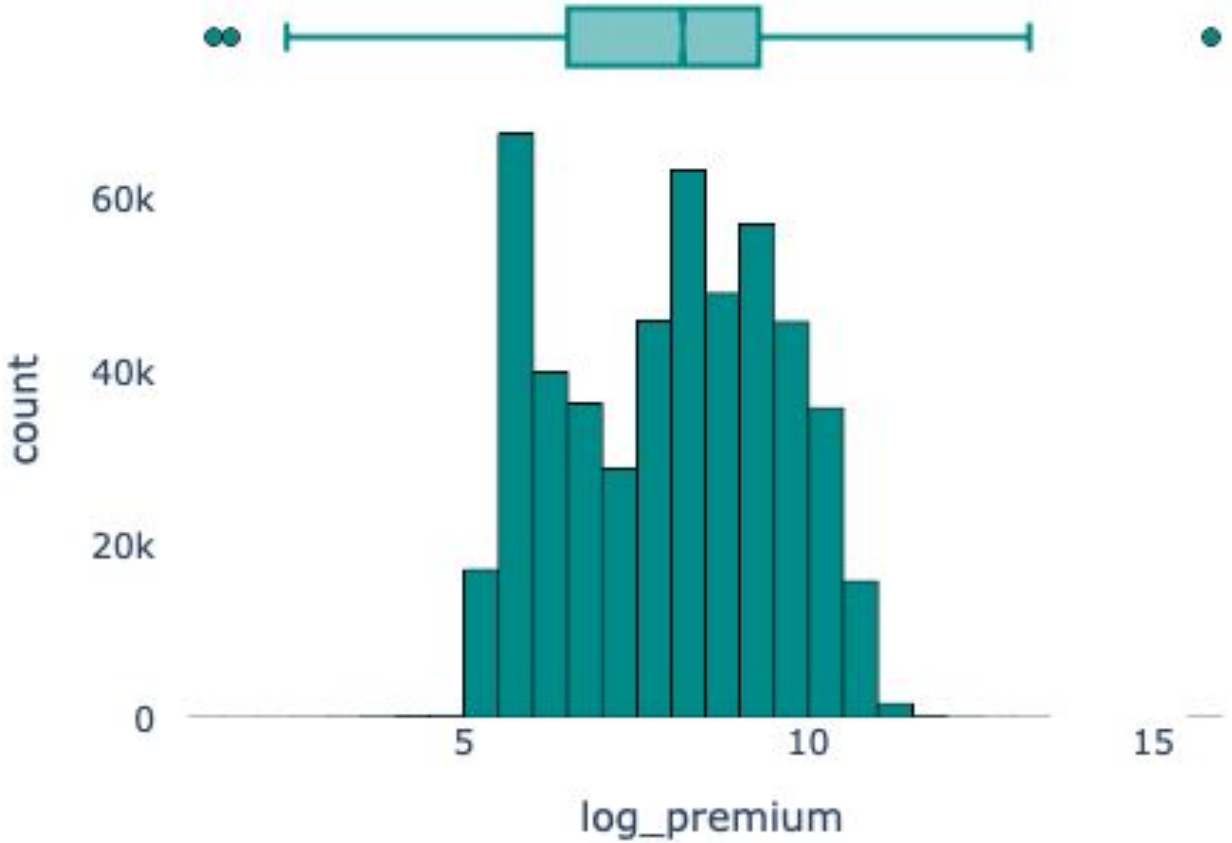


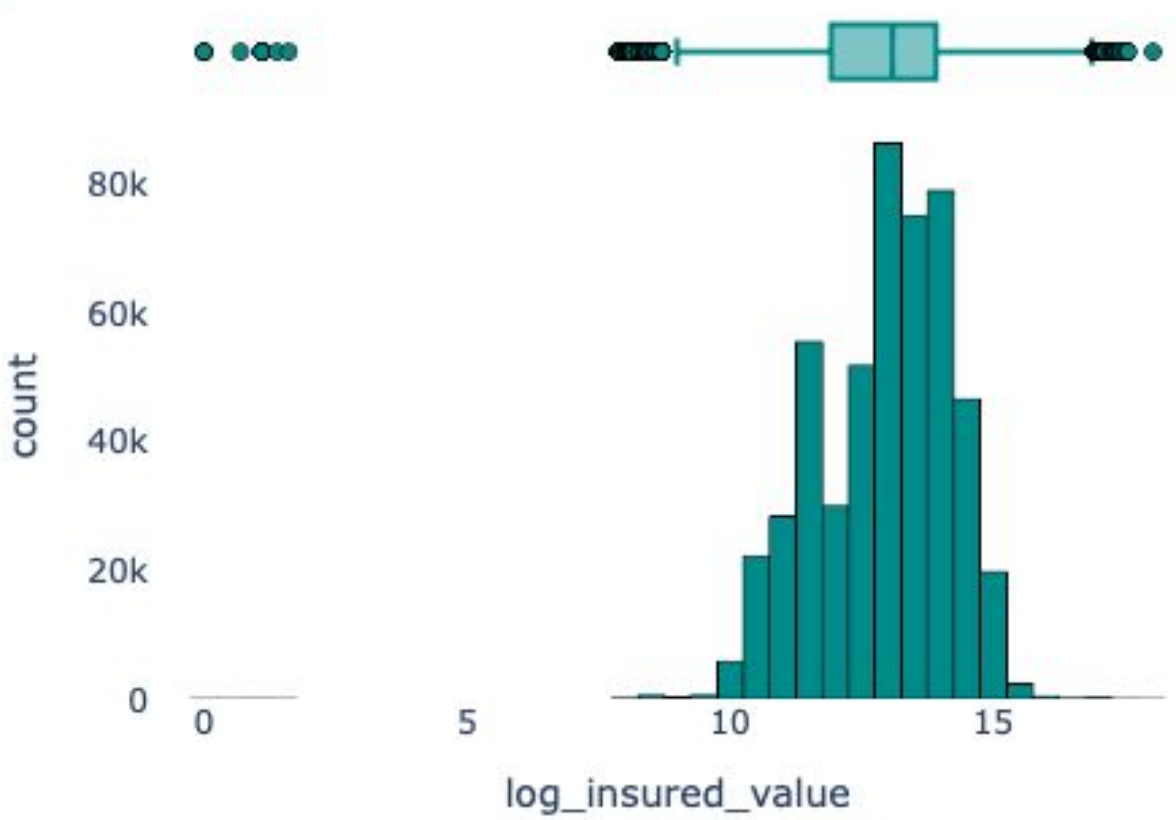




UNIVERSITY OF CALGARY

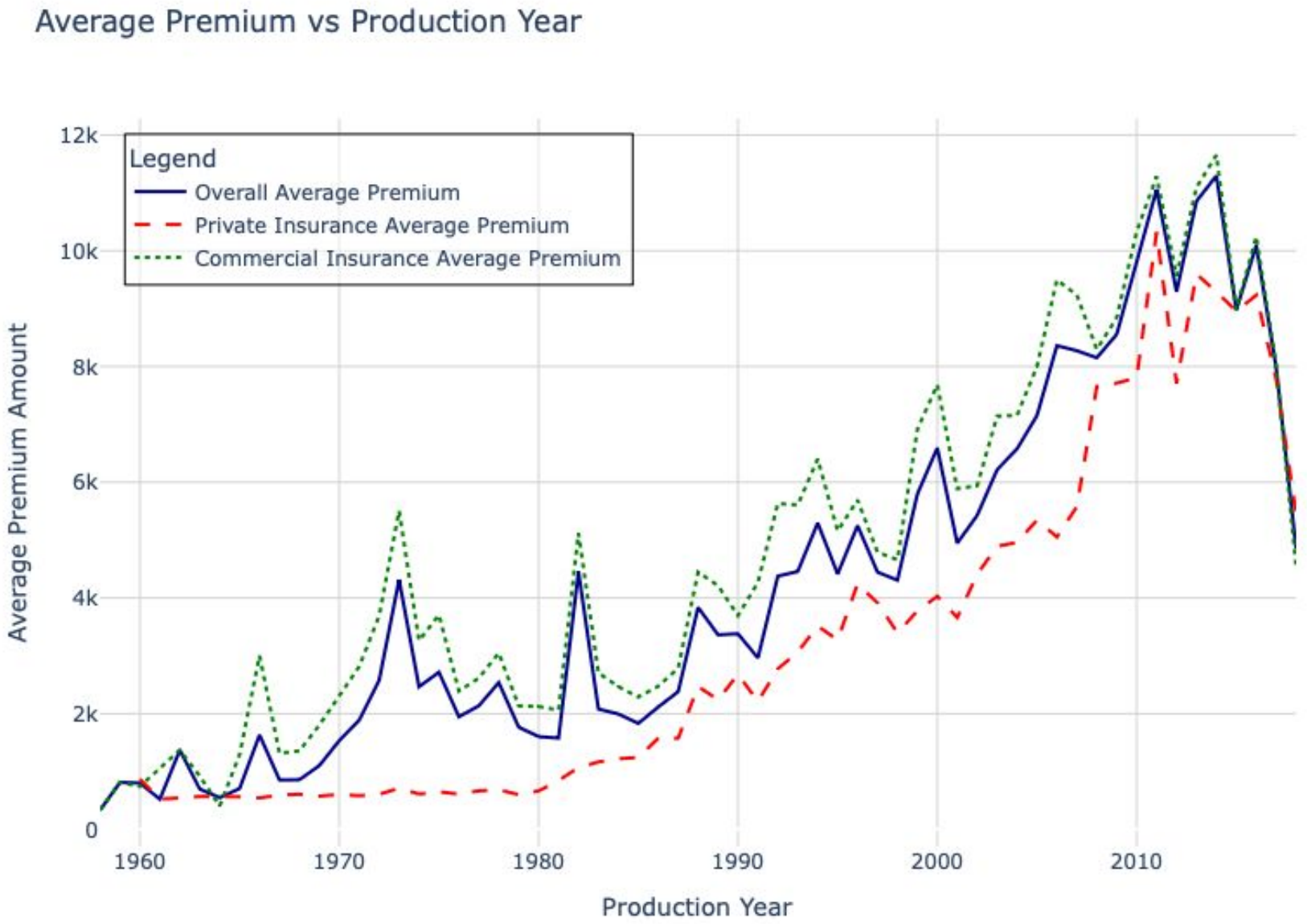
Data Visualization





UNIVERSITY OF CALGARY

Data Visualization



UNIVERSITY OF CALGARY

Analysis Insurance Premium with Random Forest Model

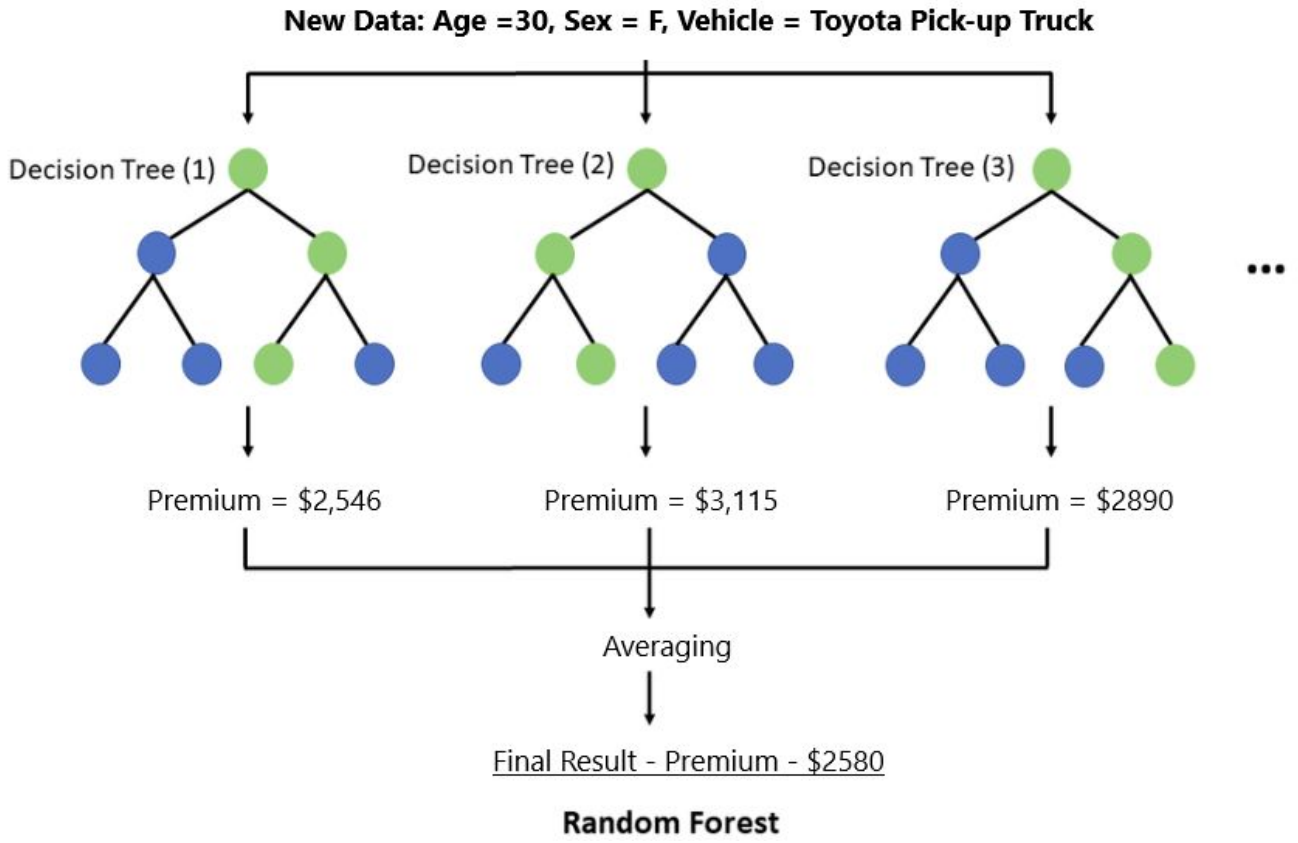


Insurance Premium and Random Forest

Vehicle Insurance Premium

- Scenario:
  - Female, Age 30
  - New Toyota Pick-up Truck
  - Insured value $300,000

- What should be the insurance premium?



UNIVERSITY OF CALGARY

Model Development Flow

1

Data Cleaning and Train-Test Split

- Missing value
- Encode categorical variable
- 80% / 20% split

2

Baseline Model

- Basic random forest model
- Default hyperparameter

3

Hyperparameter Tuning and Final Model

- Search for best hyperparameters
- Develop final model

4

Feature Importance

- Extract influencing factors of insurance premiums

UNIVERSITY OF CALGARY

Baseline Model Performance & Overfitting

- Baseline model
  - Default hyperparameters
  - No limit on max. depth, number of tree, etc
- Severe overfitting
  - Large gap between training and testing performance

Performance Metric	Training	Testing	Difference
RMSE	4,009	6,976	+73.9%
MAE	2,084	4,045	+94.0%
R^2	90.6%	71.6%	-21.0%

UNIVERSITY OF CALGARY

Model Tuning & Results

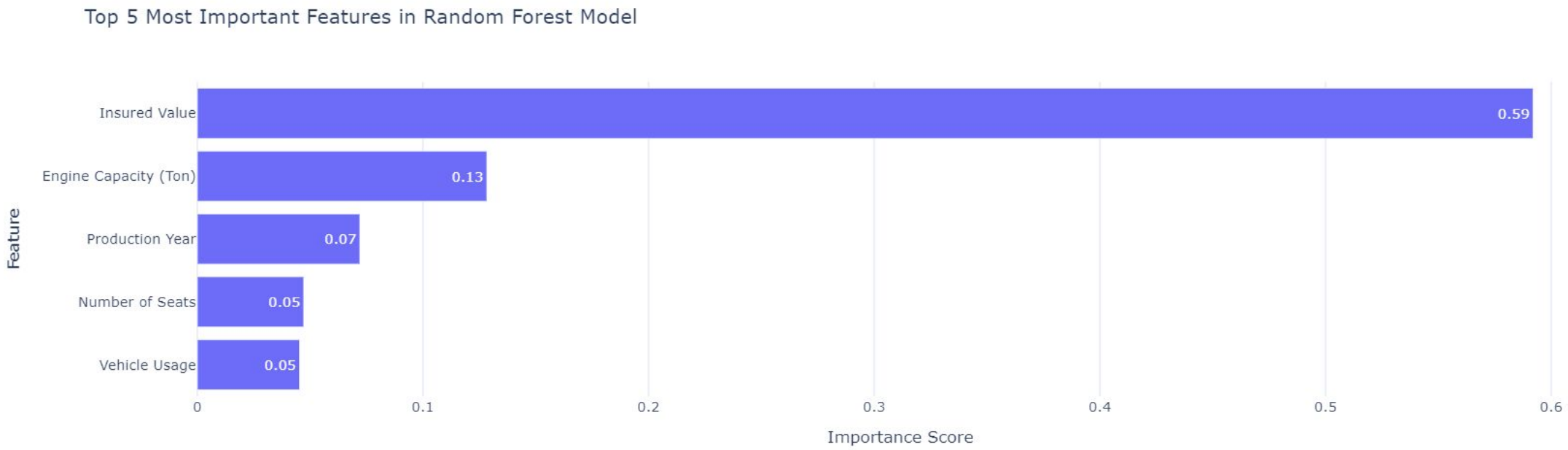
- Random Forest with Hyperparameter Tuning
  - Randomized search to identify the best hyperparameter combination
  - Cross-validation for consistent performance
  - Example of optimized hyperparameters
    - Max. feature selection = 50%
    - Max. trees = 300, max. depth = 15
    - Other settings…

Performance Metric	Baseline Model (Test)	Tuned Model (Test)	Improvement
RMSE	6,976	6,641	**+8.7%**
MAE	4,045	3,864	**+10.9%**
R^2	71.6%	74.2%	**-5.2%**

UNIVERSITY OF CALGARY

Feature Importance

- Key factors influencing car insurance premiums
  - Insured value (65.8% importance)
  - Vehicle engine capacity (11% importance)
  - Production year (6% importance)



UNIVERSITY OF CALGARY

Summary

- The importance of modeling tuning
  - Reduced overfitting (RMSE gap: 73.9% → 8.7%)
  - Better prediction accuracy

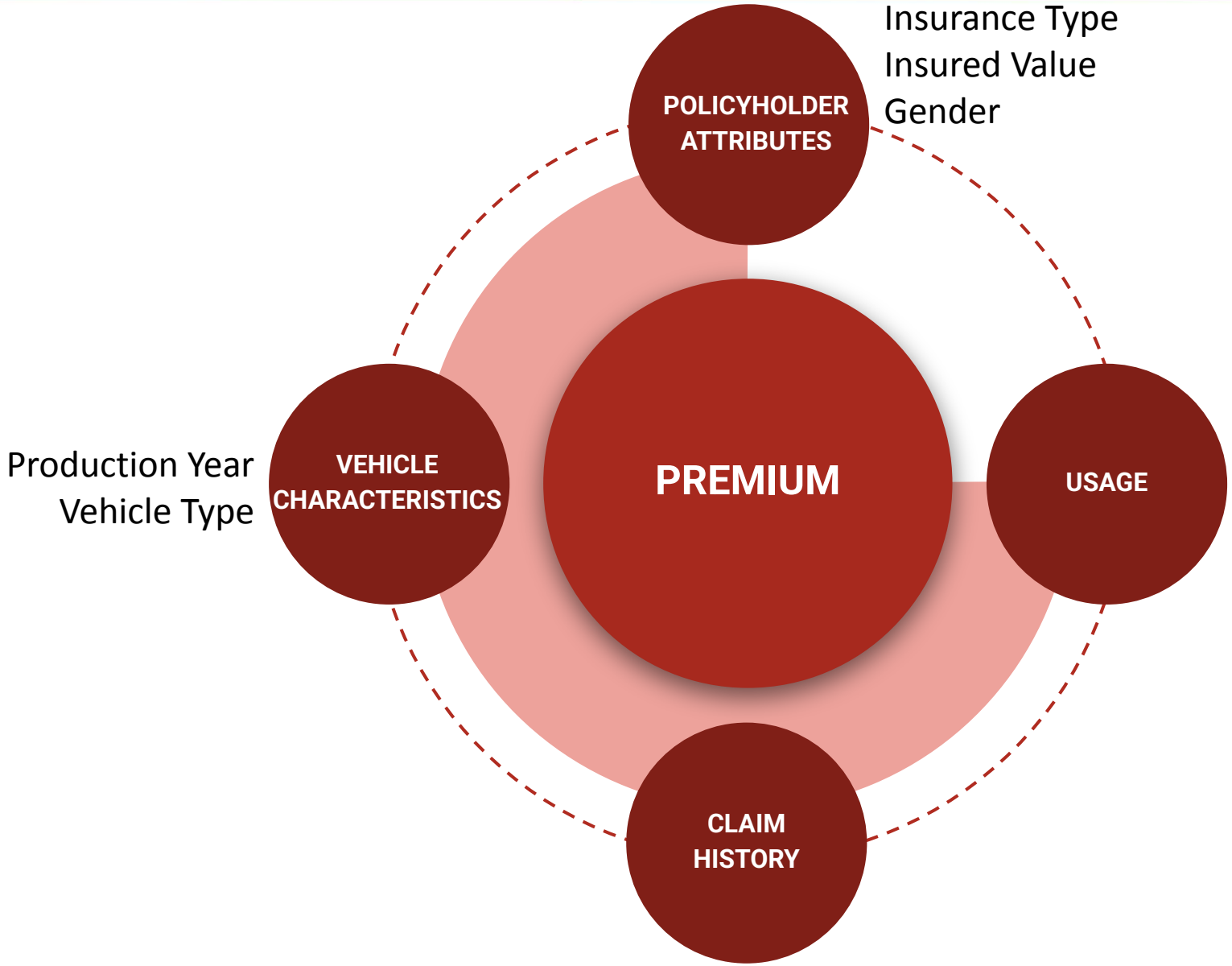
- Key factors influencing car insurance premiums
  - Insured value (65.8% importance)
  - Vehicle engine capacity (11% importance)
  - Production year (6% importance)

UNIVERSITY OF CALGARY

Analysis with Multiple Linear Regression



Response Variable & Predictors



UNIVERSITY OF CALGARY

Model Development Flow

1 — **Data Cleaning and Train-Test Split**

- Missing values
- Encode categorical variable
- 80% / 20% split

2 — **Model Fitting**

- Additive Model
- Influential Outliers
- Interaction Terms
- Polynomial Terms

3 — **MLR Assumptions Testing**

- Linearity
- Normality of Residuals
- Homoscedasticity
- Multicollinearity
- Independence of Errors

4 — **Hyperparameter Tuning**

- Lasso
- Ridge
- Elastic Net

UNIVERSITY OF CALGARY

Model Fitting

Additive Model

Metric	Training Set	Testing Set	Difference (%)
RMSE	15,100.52	8,762.19	-41.97%
MAE	4,728.66	4,648.09	-1.70%
R^2	18.37%	39.83%	+116.82%

Additive Filtered Model (drop influential outliers, use robust standard errors)

Metric	Training Set	Testing Set	Difference (%)
RMSE	8,004.63	8,585.18	+7.25%
MAE	4,330.44	4,554.07	+5.16%
R^2	46.16%	42.24%	-8.49%

UNIVERSITY OF CALGARY

Model Fitting

Interaction Model

Metric	Training Set	Testing Set	Difference (%)
RMSE	7,541.89	14,107.61	+87.06%
MAE	4,024.34	4,331.47	+7.63%
R^2	52.21%	-55.98%	-207.22%

Polynomial Model (with only insured_value squared)

Metric	Training Set	Testing Set	Difference (%)
RMSE	8,005.67	9,540.70	+19.17%
MAE	4,246.61	4,512.21	+6.25%
R^2	46.15%	28.66%	-37.90%

UNIVERSITY OF CALGARY

Hyperparameter Tuning

Model	Best Alpha	Selected Features	Train R^2	Test R^2	Train RMSE	Test RMSE	Train MAE	Test MAE
Lasso	0.01	20	45.69%	42.11%	8,039.77	8,594.47	4,321.79	4,543.91
Ridge	**1.00**	**27**	**46.16%**	**42.24%**	**8,004.64**	**8,585.17**	**4,330.44**	**4,554.07**
Elastic Net	0.01	27	45.67%	41.92%	8,041.52	8,608.54	4,338.84	4,561.25

UNIVERSITY OF CALGARY

Summary of MLR Analysis

- Used MLR to predict vehicle insurance premiums and key factors.
- Challenges: Underfitting, multicollinearity, and overfitting with interaction/polynomial terms.
- Refinements: Removed outliers, applied robust errors, and tested regularization (Lasso, Ridge, Elastic Net).
- Optimal Model: Ridge regression balanced performance, improved R^2, lowered RMSE, and retained key features.
- Key Insight: Assumption testing, feature selection, and regularization enhance insurance pricing models.

UNIVERSITY OF CALGARY

Model Comparison

Model	Train R^2	Test R^2	Train RMSE	Test RMSE	Train MAE	Test MAE
Random Forest (tuned)	**78.30%**	**74.20%**	**6,109.13**	**6,641.06**	**3,483.67**	**3,864.53**
MLR (Ridge)	46.15%	28.66%	8,005.67	9,540.70	4,246.61	4,512.21

UNIVERSITY OF CALGARY

Challenges and Limitations

UNIVERSITY OF CALGARY

Challenges and Limitations

MLR

- Challenges
  - Data quality issues
  - Model assumption violations & Interpretability
  - Computational & Optimization challenges
- Limitations
  - Assumption violations
  - Multicollinearity
  - Limited flexibility
  - Overfitting vs. Underfitting trade-off
  - Computational challenges
  - Data sensitivity

UNIVERSITY OF CALGARY

Challenges and Limitations

Random Forest

- Challenges
  - Data quality issues
  - Computational & Optimization challenges

- Limitations
  - Regional bias
    - The dataset is sourced from a single insurer. Data patterns may not generalize to other markets with different risk profiles (e.g., traffic laws, and vehicle types).

  - Limited features
    - Adding variables such as driver age, accident history, or geographic risk zones could improve model accuracy.

UNIVERSITY OF CALGARY

Recommendations

UNIVERSITY OF CALGARY

Recommendations

- **For Customers**
  - Choose lower insured value that matches your risk profile
  - Choose smaller vehicle

- **For Insurance Companies**
  - Adapt machine learning models to enhance premium pricing accuracy.
  - Enhance or automation customer information collect process to increase data quality

UNIVERSITY OF CALGARY

References

Colella, S., & Jones, H. (2023). *Machine Learning and Ratemaking: Assessing Performance of Four Popular Algorithms for Modeling Auto Insurance Pure Premium*. CAS E-Forum, Spring. https://doi.org/10.1145/2939672.2939785;citation_firstpage=785;citation_lastpage=794;citation_publication_date=2016/8/13;citation_author

Edossa Terefe. (2023). *Vehicle Insurance data. Mendeley Data*, 1. https://doi.org/10.17632/34nfrk36dt.1

Infoassurance. (2019). *Factors that affect your premium*. Infoassurance.ca. https://infoassurance.ca/en/understanding-auto-insurance/insure-your-car/factors-that-affect-your-premium/

Phillips, P., & Shang, K. (2002). *Solutions Group Applying Predictive Modeling to Auto Insurance Pricing Optimization*. PathWise®Solutions Group. https://www.casact.org/sites/default/files/2022-06/CS-4-Applying_Predictive_Modeling.pdf

Terefe, Edossa Merga. (2023). *Tree-Based Machine Learning Methods For Vehicle Insurance Claims Size Prediction*. ArXiv.org. https://arxiv.org/abs/2302.10612

The Personal. (2024). *5 factors that impact your car insurance rate*. https://doi.org/1038342/custom.css/290971f4-4d08-b626-d827-3805097c56e9

Y. Yang, A. (2025). *Insurance Premium Prediction via Gradient Tree-Boosted Tweedie Compound Poisson Models*. https://www.math.mcgill.ca/yyang/demo.html

Yu, W., Guan, G., Li, J., Wang, Q., Xie, X., Zhang, Y., Huang, Y., Yu, X., & Cui, C. (2021). *Claim Amount Forecasting and Pricing of Automobile Insurance Based on the BP Neural Network*. Complexity, 2021, 1–17. https://doi.org/10.1155/2021/6616121

UNIVERSITY OF CALGARY

Thank You!

UNIVERSITY OF CALGARY